

Agentic AI Evaluation Metrics

A practical guide to validate AI agents that plan, use tools, call APIs, execute workflows, and produce final outputs.

Recommended order: important metrics first, simple validation questions, evaluation flow, then detailed explanation.

1. Most Important Agentic AI Metrics to Remember

Agentic AI evaluation is different from normal LLM evaluation. We do not only check the final answer. We also check whether the agent followed the right steps, selected the right tools, passed correct inputs, respected permissions, and created an auditable workflow.

Metric	What it checks	Simple example
Task Success Rate	Did the agent complete the user goal?	Generated the correct compliance report.
Tool Selection Accuracy	Did the agent choose the correct tool or API?	Used database search for customer records, not web search.
Parameter Accuracy	Did the agent pass the right inputs to tools?	Correct customer ID, date range, product type.
Trajectory Accuracy	Did the agent follow the correct sequence of steps?	Retrieve data -> validate -> analyze -> draft report.
Workflow Compliance	Did the agent follow business process and policy?	Did not skip KYC or approval checks.
Human-in-the-Loop Accuracy	Did it ask for human approval when needed?	Asked manager approval before final loan decision.
Safety and Permission Compliance	Did it avoid unsafe or unauthorized actions?	Did not send email or update record without permission.
Auditability	Can we review what the agent did and why?	Logs show plan, tool calls, inputs, outputs, and final action.

2. Agentic AI Metrics with Simple Questions

Metric	Simple validation question
Task Success Rate	Did the agent finish the task correctly?
Goal Alignment	Did the agent solve the actual user need, not just produce text?
Plan Quality	Was the plan logical, complete, and efficient?
Step Relevance	Were the steps useful, or did the agent do unnecessary work?
Tool Selection Accuracy	Did it choose the correct tool for each step?
Tool Call Success Rate	Did the tool/API calls execute successfully?
Parameter Accuracy	Were the tool inputs correct and complete?
Error Recovery	Did the agent recover when a tool failed or returned missing data?
Trajectory Accuracy	Did it follow the right path from start to finish?
Workflow Compliance	Did it follow required enterprise/banking process rules?
Human Approval Trigger	Did it request approval for sensitive actions?
Permission Compliance	Did it avoid actions outside user/system permission?
Data Privacy Compliance	Did it protect confidential data and PII?
Auditability	Can we trace decisions, tool calls, and outputs later?

Simple rule: For an agent, a correct final answer is not enough. The path must also be correct, safe, approved, and auditable.

3. Simple Agentic AI Evaluation Flow

Step	What to evaluate	Metrics to use
Step 1: Evaluate the user goal	Did the agent understand the task and expected outcome?	Goal alignment, task success criteria
Step 2: Evaluate the plan	Did it create a logical and complete plan before acting?	Plan quality, step relevance, missing-step rate
Step 3: Evaluate tool usage	Did it choose the right tools and call them correctly?	Tool selection accuracy, parameter accuracy, tool success rate
Step 4: Evaluate workflow path	Did it follow the right sequence and enterprise process?	Trajectory accuracy, workflow compliance, order correctness
Step 5: Evaluate final output	Is the final response/report/action correct and useful?	Answer correctness, completeness, relevance
Step 6: Evaluate safety controls	Did it protect data and avoid unauthorized actions?	Privacy compliance, permission compliance, human approval accuracy
Step 7: Evaluate operations	Is the system reliable, cost-effective, and traceable?	Latency, cost per task, failure rate, auditability

Banking example: A loan-review agent should not only produce a recommendation. It must check KYC, credit score, income documents, policy rules, fraud signals, exceptions, and human approval rules in the correct order.

4. Detailed Explanation of Agentic AI Metrics

Task Success Rate

This checks whether the agent completed the requested task correctly. For example, if the user asks the agent to prepare a risk summary, the output should contain the correct issue, impact, risk level, recommendation, and next action.

Goal Alignment

This checks whether the agent solved the real business goal. Sometimes an agent gives a polished answer but misses the actual requirement. For example, the user asks for high-risk cases, but the agent summarizes all cases.

Plan Quality

This checks whether the agent created a good step-by-step plan. A good plan is clear, complete, and not unnecessarily complex. In enterprise use cases, planning is important because actions may affect customers, records, reports, or compliance.

Step Relevance

This checks whether each step taken by the agent was useful. Unnecessary steps increase cost, latency, and risk. For example, an agent should not call five tools when one internal database query is enough.

Tool Selection Accuracy

This checks whether the agent selected the correct tool. For example, to calculate numbers it should use a calculator or code tool; to read internal policy it should use the document search tool; to create an event it should use the calendar tool.

Parameter Accuracy

This checks whether the agent passed correct values to tools. Even the right tool can fail if the wrong date range, customer ID, file name, branch code, or product category is passed.

Tool Call Success Rate

This measures how many tool calls completed successfully. Failed API calls, bad permissions, missing files, and invalid parameters reduce reliability.

Error Recovery

This checks whether the agent can handle failures. For example, if one data source is unavailable, the agent should retry, use an approved fallback, or clearly explain the limitation instead of inventing information.

Trajectory Accuracy

Trajectory means the full path the agent followed. This metric checks whether the agent followed the right sequence of decisions and tool calls. This is very important because an agent can reach a correct-looking answer through an unsafe or non-compliant path.

Workflow Compliance

This checks whether the agent followed required business rules. In banking, it may need to check KYC, AML, sanctions, credit policy, approval limits, and audit requirements before taking action.

Human-in-the-Loop Accuracy

This checks whether the agent asks for human approval at the right time. Sensitive actions such as approving a loan, sending a customer email, changing a record, or escalating a case should often require approval.

Permission Compliance

This checks whether the agent respects access control and authority. The agent should not read restricted data, modify records, send messages, or trigger workflows unless the user and system permissions allow it.

Data Privacy Compliance

This checks whether the agent protects sensitive information such as account numbers, customer identifiers, financial details, personal data, internal policies, and confidential documents.

Output Correctness

This checks whether the final output is accurate. For example, a case-summary agent should not mix customer names, wrong dates, incorrect account details, or unsupported risk levels.

Completeness

This checks whether the final output includes all required points. A compliance report may need issue, evidence, risk rating, policy reference, recommended action, owner, and deadline.

Latency

This measures how long the agent takes to complete the task. Agents may make multiple tool calls, so latency can become high if planning and tool use are not optimized.

Cost per Task

This measures total cost for one completed task, including LLM tokens, embeddings, database calls, API calls, and tool execution cost.

Auditability

This checks whether the agent keeps a clear record of the user request, plan, tool calls, tool inputs, tool outputs, decisions, approvals, and final response. This is critical for regulated industries.

5. Practical Banking Validation Checklist

Validation area	Checklist question
Goal	Did the agent understand the exact banking task?
Data	Did it use the right approved data sources?
Tools	Did it choose the right internal tools/APIs?
Inputs	Did it pass correct parameters such as customer ID, date, branch, product, or case number?
Workflow	Did it follow the correct KYC, AML, risk, approval, and escalation flow?
Safety	Did it avoid exposing PII or confidential data?
Approval	Did it ask for human approval before sensitive actions?
Final output	Was the answer/report/action correct, complete, and clear?
Audit	Can all steps and decisions be reviewed later?

Most practical teaching summary: Evaluate an AI agent on the final result, the tools it used, the sequence of steps it followed, the business rules it respected, and the audit trail it produced.